Indian Institute of Technology, Bhilai

CSP203: Software Tools & Technologies Laboratory

Lab Activity - 13

Full Marks: 30 *Deadline: 5:30 PM, 09-10-2025*

Submission Instructions:

1) Answer all the questions.
2) You should prepare a zip file which will contain all the .py, and .pdf files of the task in their respective folders.
3) Each Gnuplot script file should be placed in its respective task folder (e.g., Task1, Task2..).
4) Your zip file should be named as <Roll No>_<Name>_<Lab Activity No>.zip
5) Example: B24CS055_Shareen_13.zip
6) **Plagiarism is strictly prohibited.**

Objectives:

1) Understand the Basics of Plotting.
2) Learn how to create simple line plots, bar charts, scatter plots, histograms, and pie charts using Python lists.
3) Customize Visual Elements.
4) Gain skills to modify colors, markers, line styles, bar widths, transparency, and figure sizes to enhance visual clarity.
5) Labeling and Annotation.
6) Learn to add axis labels, titles, legends, and custom tick marks to make plots informative and publication-ready.
7) Compare Multiple Data Series.

8) Understand how to plot multiple datasets in a single graph (e.g., two scatter plots) to analyze relationships and trends effectively.
9) Data Representation and Interpretation.
10) Develop the ability to choose the appropriate plot type for a dataset and interpret visual insights for decision-making and reporting.

Task1: [10 Marks]

Problem Statement:

A student has tracked the number of hours spent on different subjects during the week. Your task is to create a pie chart that visually represents the percentage of time spent on each subject, and to save the output as a PDF file for reporting purposes.
This chart will help analyze how the student allocates their study time and identify which subjects receive more or less attention.

Data Provided:

You are given the list of subjects and the corresponding number of study hours:

```
subjects = ['Math', 'Science', 'English', 'History', 'Computer Science']
study_hours = [10, 8, 6, 4, 12]
colors = ['#ff9999', '#66b3ff', '#99ff99', '#ffcc99', '#c2c2f0']
```

Your Tasks:

1. Import matplotlib.pyplot and any other necessary libraries.
2. Create a pie chart using the provided data:
3. Use subjects for labels.
4. Use study_hours for slice sizes.
5. Use colors for slice coloring.
6. Set up the chart with the following configurations:
7. startangle = 140: Rotates the start of the pie chart.
8. shadow = True: Adds a shadow for depth.
9. explode = (0, 0, 0, 0, 0.1): Slightly highlight Computer Science by popping it out.
10. autopct = '%1.1f%%': Show percentage values inside slices.
11. radius = 1.1: Slightly increase the chart's size.
12. figsize = (6, 6): Make the figure square.
13. Add:

a. A legend matching subjects to colors.
b. A title: "Weekly Study Time Allocation".

Output Requirement:

Save the figure as a PDF file named:
study_time_pie_chart.pdf

Expected Output:

A pie chart saved as a PDF that:

1. Visually displays the percentage of time spent on each subject.
2. Clearly highlights Computer Science as the most studied subject.
3. Includes a color-coded legend and a descriptive title.

Task 2: [10 Marks]

A retail company sells several types of products and has recorded their total monthly sales (in units). Your task is to create a bar plot to visualize the number of units sold for each product. This will help management compare performance across product categories and plan inventory accordingly.

Given Data:

You are provided with the following product sales data:
products = ['Laptops', 'Smartphones', 'Headphones', 'Monitors', 'Keyboards']
units_sold = [150, 300, 250, 130, 180]

Tasks:

1. Create a vertical bar plot using Matplotlib to display product-wise sales.
2. Customize the chart with the following:
3. Use different colors for each bar.
4. Set a suitable figure size (e.g., 10×6 inches).
5. Add axis labels:
    a. X-axis: "Product Category"
    b. Y-axis: "Units Sold"
6. Add a title: "Monthly Sales Distribution of Products"
7. Use Times New Roman font for labels and title.
8. Display value labels above each bar showing the exact number of units sold.
9. Add gridlines on the Y-axis for better readability.
10. Save the plot as a PDF file named "sales_barplot.pdf".

Expected Output:

1. A clean, colorful bar chart where each bar corresponds to a product.
2. Values should be labeled above the bars.
3. The plot should be publication-ready with proper fonts, labels, and spacing.

Task 3: [10 Marks]

Problem Statement:

You are studying the relationship between students' heights and weights. The students belong to two groups: Group 1 (Male) and Group 2 (Female). The goal is to visualize whether taller students tend to weigh more and compare the two groups using a scatter plot.

Data Provided:

Male students (Height in cm, Weight in kg)
male_heights = [150, 155, 160, 162, 165, 168, 170, 172, 175, 178]
male_weights = [50, 53, 55, 58, 60, 62, 65, 67, 70, 73]

—---

Female students (Height in cm, Weight in kg)
female_heights = [148, 152, 157, 160, 163, 165, 168, 170, 173, 175]
female_weights = [48, 50, 53, 55, 58, 60, 62, 64, 67, 69]

Tasks:

1. Define the data as Python lists.
2. Plot two scatter plots on the same figure: one for male students and one for female students.
3. Use distinct markers (* for males, o for females) and colors for each group.
4. Customize the size of the points.
5. Label the X-axis as "Height (cm)" and Y-axis as "Weight (kg)".
6. Add a suitable title: "Height vs Weight of Students by Group".
7. Add gridlines and a legend to distinguish the two groups.
8. Save the plot as a PDF file named "height_scatterplot.pdf".

Expected Output:

4. A clean, colorful scatter plot where each bar corresponds to a product.
5. Values should be labeled above the left corner of the plot.
6. The plot should be publication-ready with proper fonts, labels, and spacing.